



Technical Service Bulletin: tsb100892

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New Intake and Exhaust Valves

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document announces the release of new intake and exhaust valves for use on all QSK45, all QSK60 (including natural gas), and QSK19 engines (other than QSK19 MCRS). The new valves have a constant stem diameter to the end of the valve stem tip to improve robustness.

Due to the increased diameter of the valve stem tip, the new valves require new spring retainers, collets, and crossheads. The cylinder head casting, valve guides, rotators, springs, and rocker levers remain unchanged. New part numbers have also been released for all cylinder head assemblies affected by this change. Reference Table 1 below.

To assist in identifying the new parts, the new crosshead has a circle stamped on the side opposite to the part number and the new collets have a copper color appearance. The new spring retainer has a larger inner profile. To identify the new from the old retainer, measure the inner diameter at a distance of 2.3 mm [0.09 in] down from the top of the retainer. If an old retainer is used, the new collets will **not** sit flush to the top of the retainer. Reference the illustrations below for new crosshead, collet, and spring retainer identification and measurement.

It is essential that all the valves are of the same design level on any one cylinder head, but it is acceptable to have cylinder heads of each design level mounted on the same engine.

The new cylinder head assemblies went in to production for QSK45/60 engines on 27 October 2009, ESN first: 33179494; and for QSK19 engines on 5 November 2009, ESN first: 37241838.

Note : This is a product improvement and is **not** subject to TRP or campaign actions.

Note : There are different intake and exhaust valve part numbers, depending on the engine model. Make sure the correct valve part number is identified before ordering parts.

Note : The new crosshead **must only** be used with the new increased diameter valve stem tip. Do **not** install the new crosshead to the old style thinner valve stem tip valves. Engine damage can possibly occur.

Note : The new intake valves utilize a nitride hardening process known as Sursulf. Sursulf valves can **not** be ground. Grinding will remove a significant portion of the hardened layer. Ground valves will be susceptible to 'beat in' wear and premature

malfunction. Sursulf valves have a darker appearance, when compared to the previous design valve.

Table 1: Parts Affected

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Part Description	Old Part Number	New Part Number
Collets	3081534	3641537
Crosshead	3640458	3641535
Exhaust Valve	4065481	3641541
Exhaust Valve	3090469	3641542
Exhaust Valve	4097624	3641543
Exhaust Valve	4951663	3641599
Intake Valve	3171769	3641538
Intake Valve	3635051	3641539
Intake Valve	4016438	3641540
Retainer	3331752	3641536

Table 2: Cylinder Head Assemblies Affected		
Engine Model(s)	Old Part Number	New Part Number
QSKTA60	2866707	3641558
QSK60-G	3636354	3641561
QSK45-C, QSK45-G, QSK60-C, QSK60-G	3638747	3641562
QSK60-M, QSK60-D(M)	4086077	3641564
QSK60-M, QSK60-D(M)	4098444	3641565
QSKTA60-ME, QSKTA60-GE, QSKTA60-CE, QSKTTA60-CE	4951826	3641567
GQSK60-G	4968967	3641570
GQSK60-G	4999071	3641571
QSK19-M	3331729	3641559
GQTA19-G, GQTA19-C	4922192	3641566
QTA19-C	4953480	3641569
QTA-19C, QTA19-C	4953479	3641568
QTA19-R	3408281	3641560

The following service kits contain 1 valve and 2 collets.

Table 3: New Valve and Collet Kits		
Old Part Number	New Part Number	Contents
4955384	2881810	2x Collets (3641537) and 1x Intake Valve (3641538)
4025070	2881811	2x Collets (3641537) and 1x Exhaust Valve (3641541)
3804892	2881812	2x Collets (3641537) and 1x Exhaust Valve (3641542)
4089504	2881813	2x Collets (3641537) and 1x Exhaust Valve (3641543)
4955953	2881814	2x Collets (3641537) and 1x Exhaust Valve (3641599)
3800955	2881815	2x Collets (3641537) and 1x Intake Valve (3641540)
3641822	2881822	2x Collets (3641537) and 1x Intake Valve (3641539)

Three new cylinder head overhaul kits are available which contain the new valves, collets, retainers, and crossheads. Reference Tables 4, 5 and 6.

The remove and install procedures for the cylinder heads and various components have **not** changed. Reference Procedure 002-004 (Cylinder Head) in the relevant service manual for your engine: QSK45/60 – Service Manual 4021530, QSK19 – Service Manual 4021592.

Table 4: Kit 1 - Part Number 2881819		
Quantity	Part Description	New Part Number
2	Intake Valve	3641540
2	Exhaust Valve	3641541
8	Collets	3641537
4	Retainers	3641536
2	Crossheads	3641535

Table 5: Kit 2 - Part Number 2881820		
Quantity	Part Description	New Part Number
2	Intake Valve	3641538
2	Exhaust Valve	3641541
8	Collets	3641537
4	Retainers	3641536
2	Crossheads	3641535

Table 6: Kit 3 - Part Number 2881821

Quantity	Part Description	New Part Number
2	Intake Valve	3641540
2	Exhaust Valve	3641542
8	Collets	3641537
4	Retainers	3641536
2	Crossheads	3641535

The three new complete cylinder head overhaul kits shown in Tables 4, 5 and 6 will be applicable to the following cylinder head assemblies and engine models; reference Tables 7, 8 and 9.

Note : Not all cylinder head assemblies and engine models are covered by these three new cylinder head overhaul kits. If a cylinder head is not highlighted in Tables 7, 8 or 9 below, the individual parts will need to be ordered for a cylinder head overhaul; reference Tables 1 and 3 above for individual parts listings.

Table 7: Kit 1, Part Number 2881819, Where Used

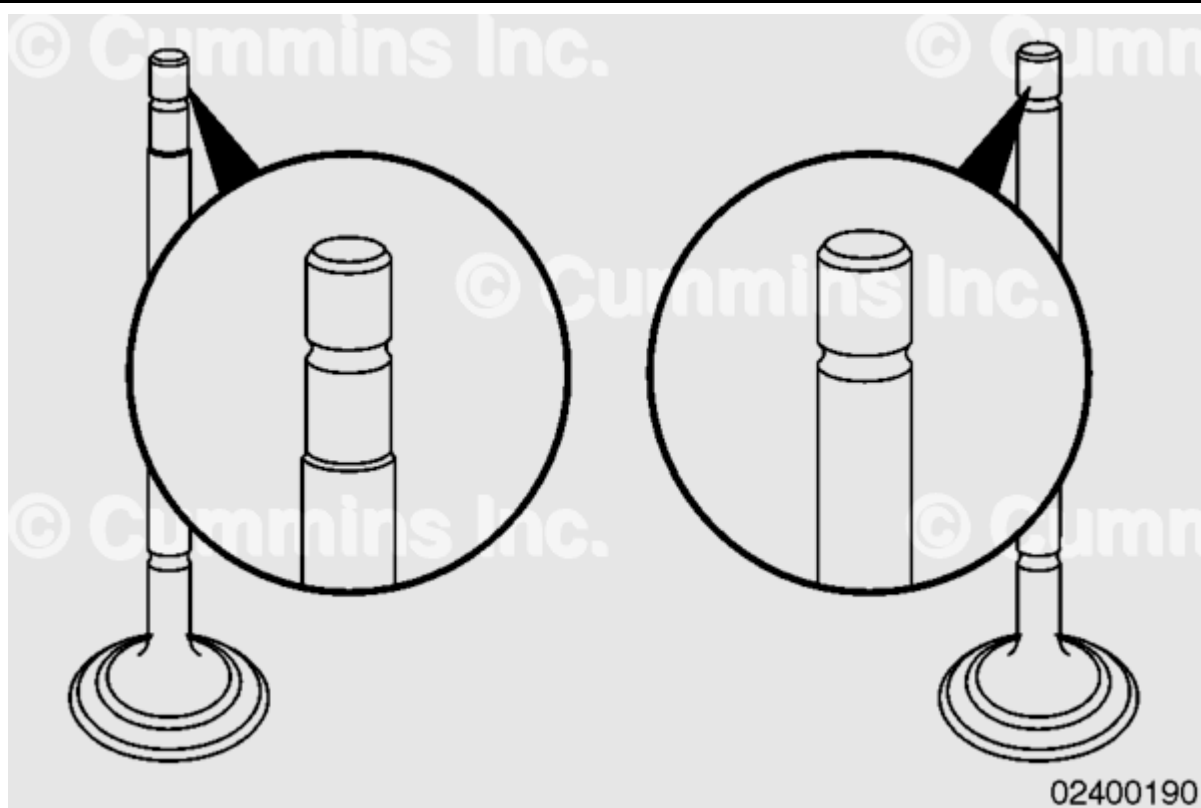
Engine Models	Old Cylinder Head Part Number	New Cylinder Head Part Number
QSKTA60	2866707	3641558
QSK60-M, QSK60-D(M)	4065581	3641564
QSK60-M, QSK60-D(M)	4080224	3641564
QSK60-M, QSK60-D(M)	4080358	3641564
QSK60-M, QSK60-D(M)	4086077	3641564
QSK60-M, QSK60-D(M)	4100537	3641564
QSK60-M, QSK60-D(M)	4097849	3641565
QSK60-M, QSK60-D(M)	4098444	3641565
QSKTA60-ME, QSKTA60-GE, QSKTA60-CE, QSKTTA60-CE	4917123	3641567
QSKTA60-ME, QSKTA60-GE, QSKTA60-CE, QSKTTA60-CE	4951826	3641567
QSK60-M, QSK60-D(M)	4001093	3641564
QSK60-M, QSK60-D(M)	4016795	3641564
QSK60-M, QSK60-D(M)	4086077	3641564

Table 8: Kit 2, Part Number 2881820, Where Used

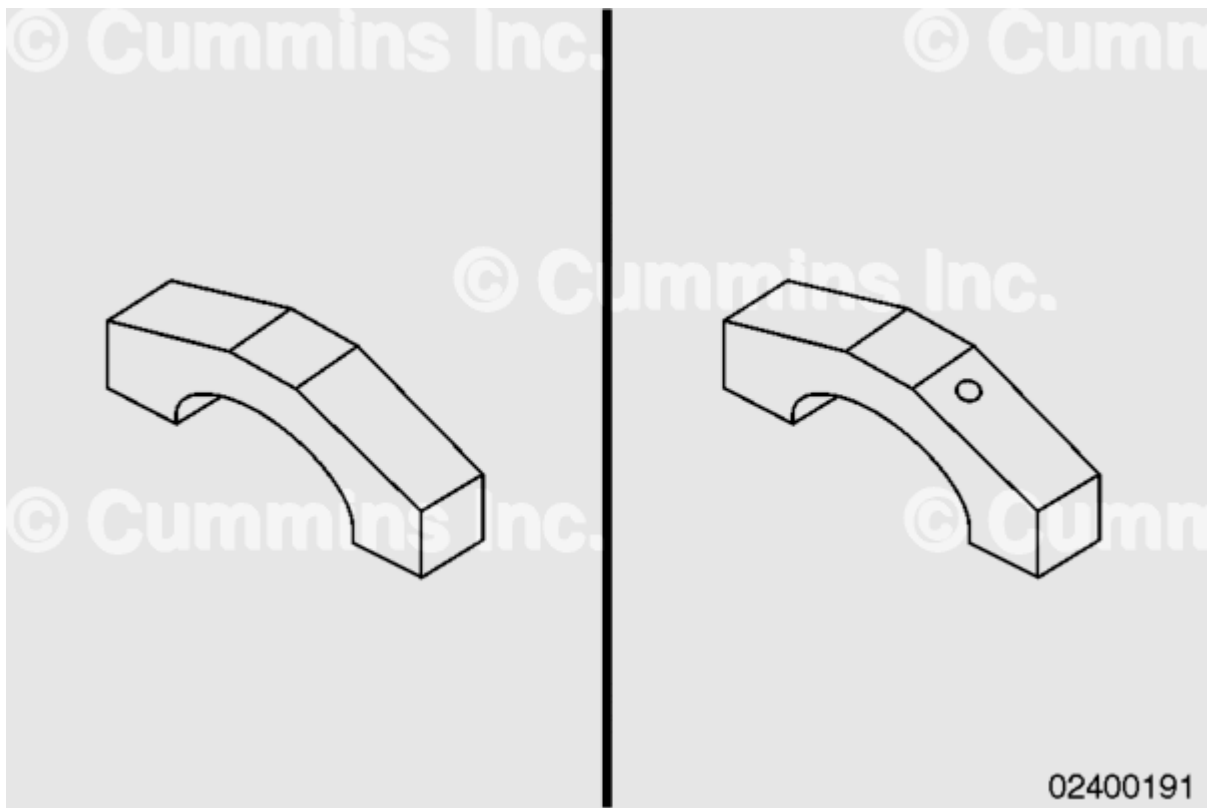
Engine Models	Old Cylinder Head Part Number	New Cylinder Head Part Number
QSK45-C, QSK45-G, QSK60-C, QSK60-G	3638747	3641562

Table 9: Kit 3, Part Number 2881821, Where Used

Engine Models	Old Cylinder Head Part Number	New Cylinder Head Part Number
QSK19-M	3090369	3641559
QSK19-M	3170137	3641559
QSK19-M	3331729	3641559
QTA19-R	3408281	3641560

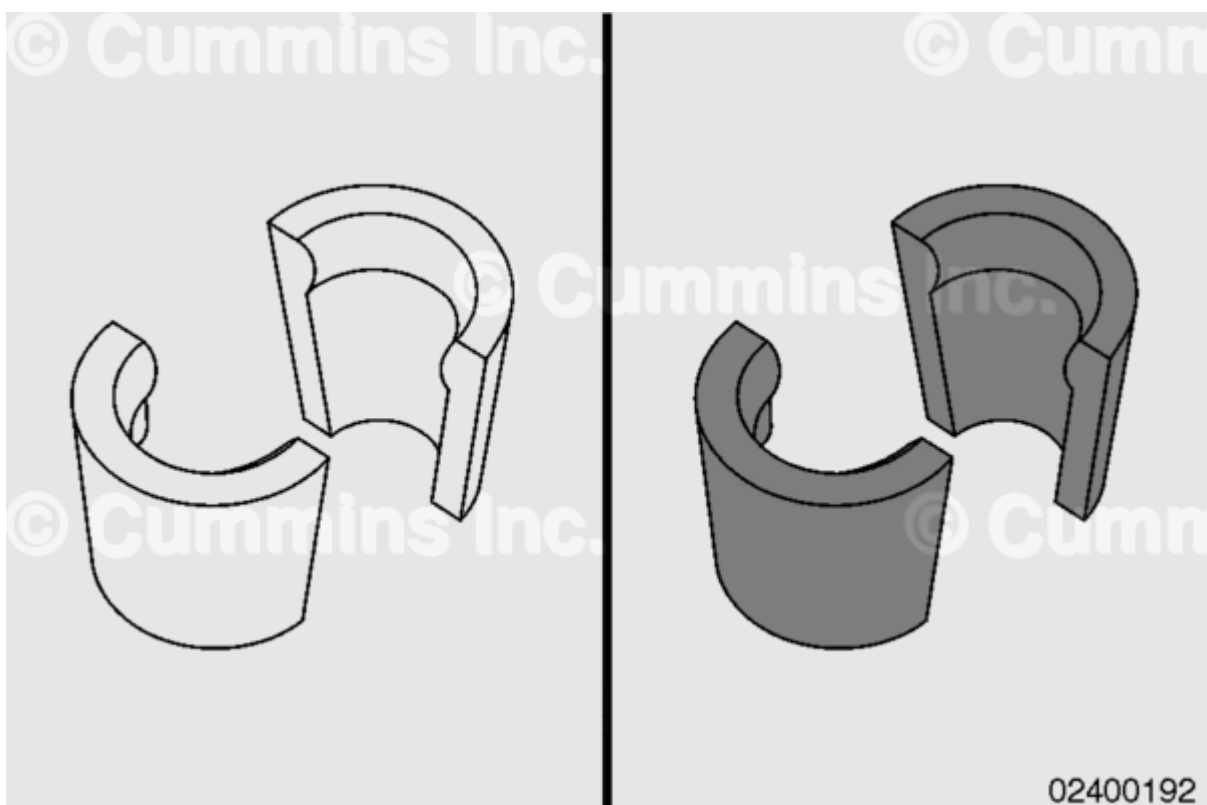
**Old (left) and New (right) Intake and Exhaust Valve**

1. Thicker valve tip on the new valve
2. Collet groove.



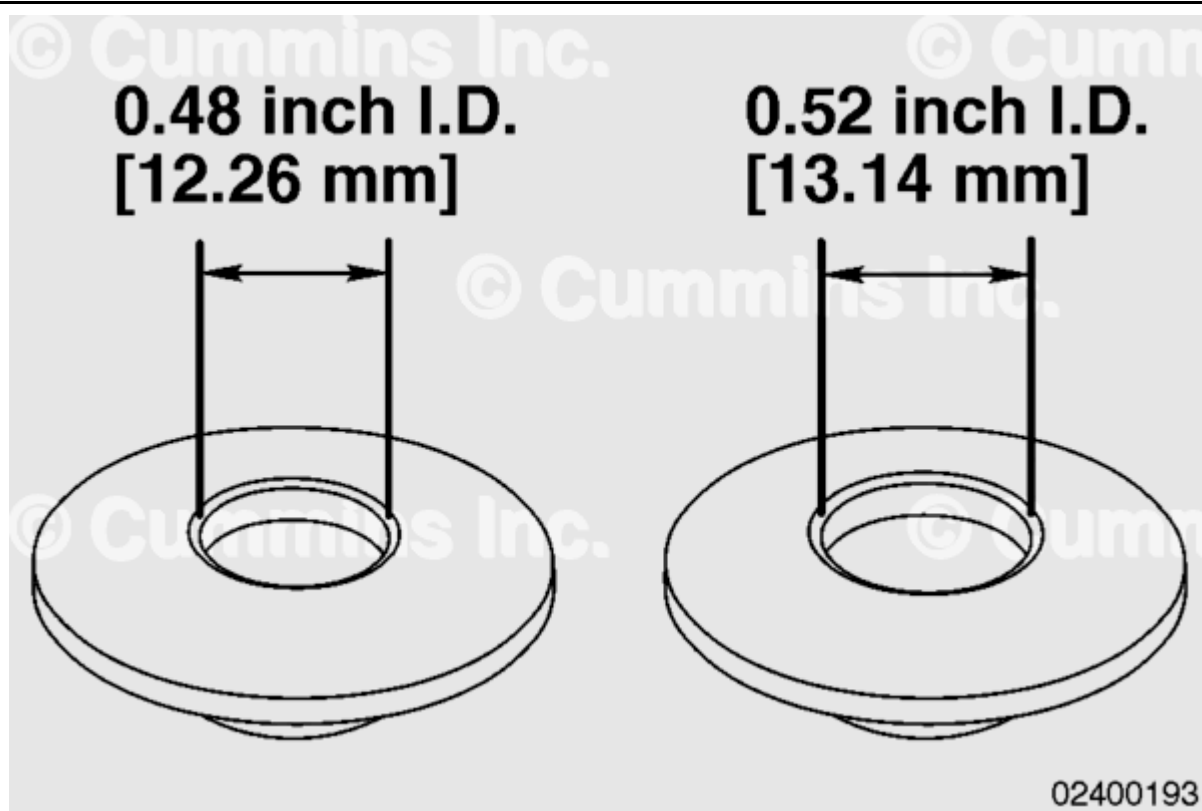
Old (left) and New (right) Crosshead

1. Circle stamped on new crosshead.



Old (left) and New (right) Collets

Note : New collets are copper in color.



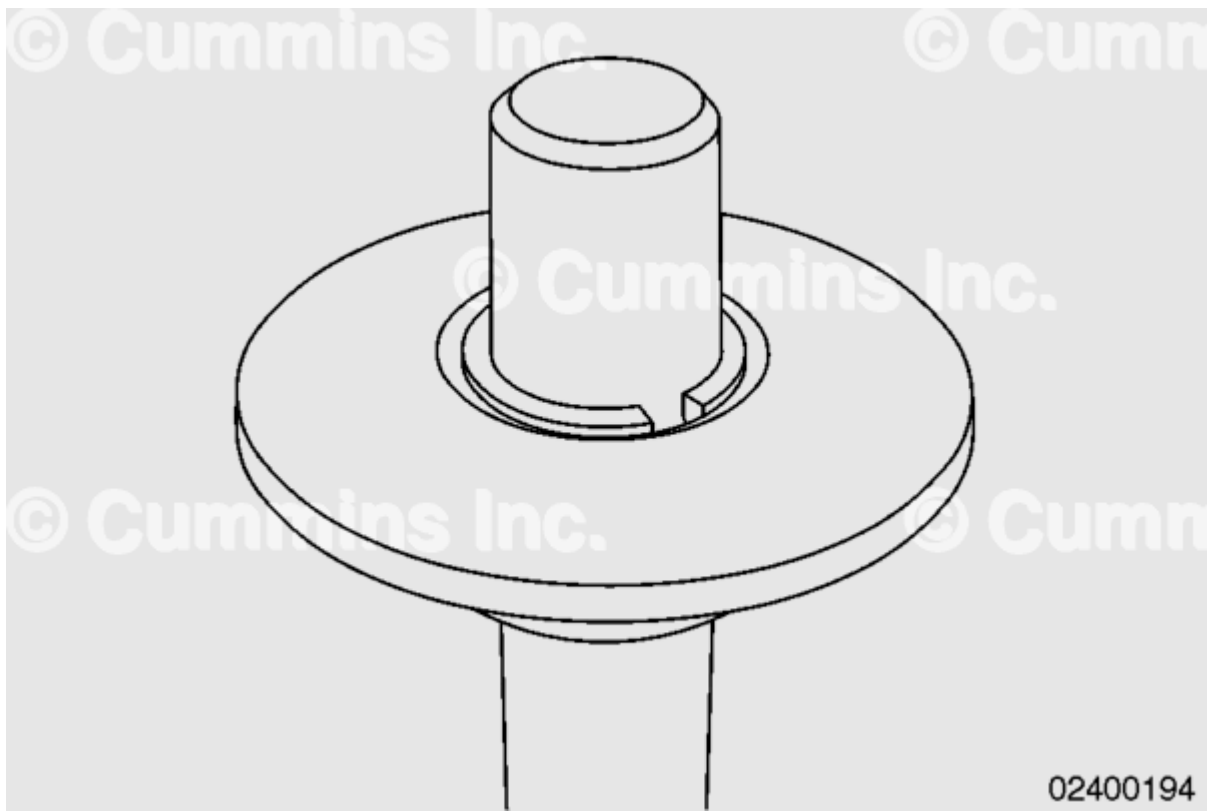
Old (left) and New (right) Retainers

Note : The new retainers have a larger internal profile. If an old retainer is used, the new collets will not sit flush to the top of the retainer.

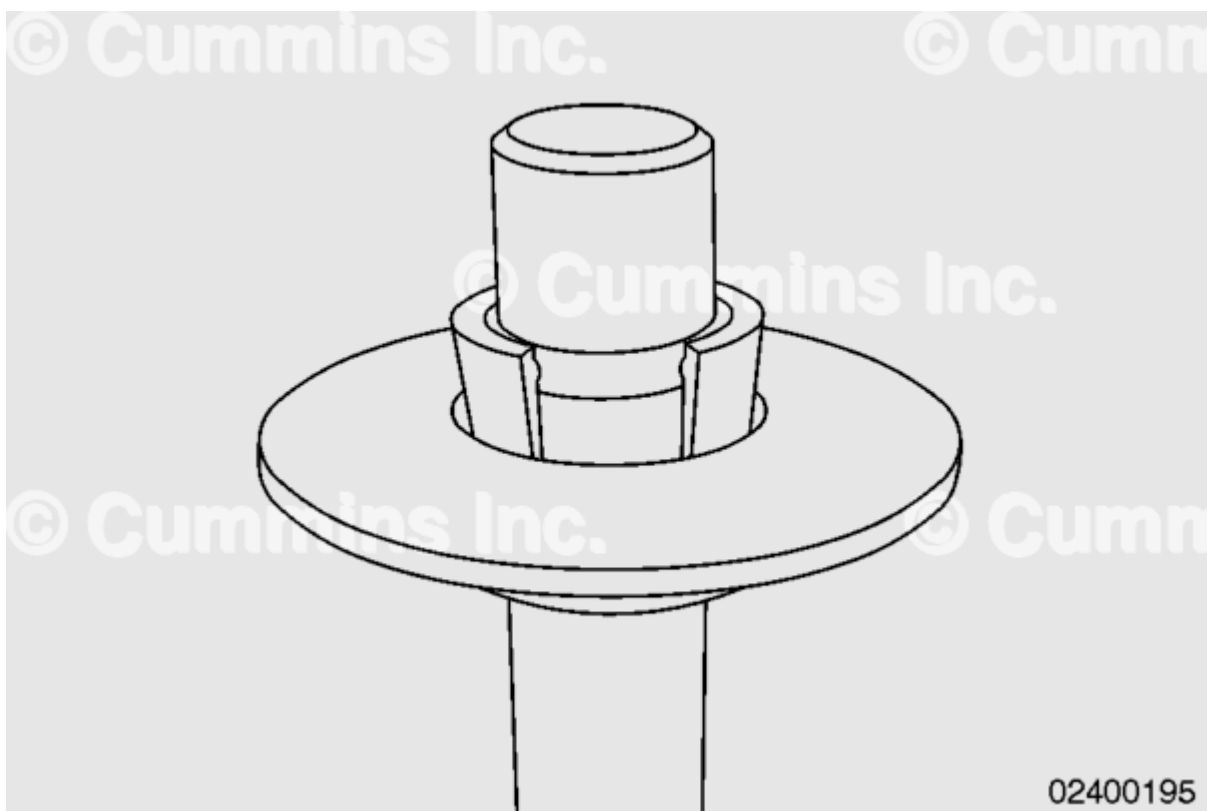
Measure the inside diameter at a distance of 2.3 mm [0.09 in] down from the top of the retainer.

Old retainer inside diameter at 2.3 mm [0.09 in] down from top: 12.26 mm [0.48 in]

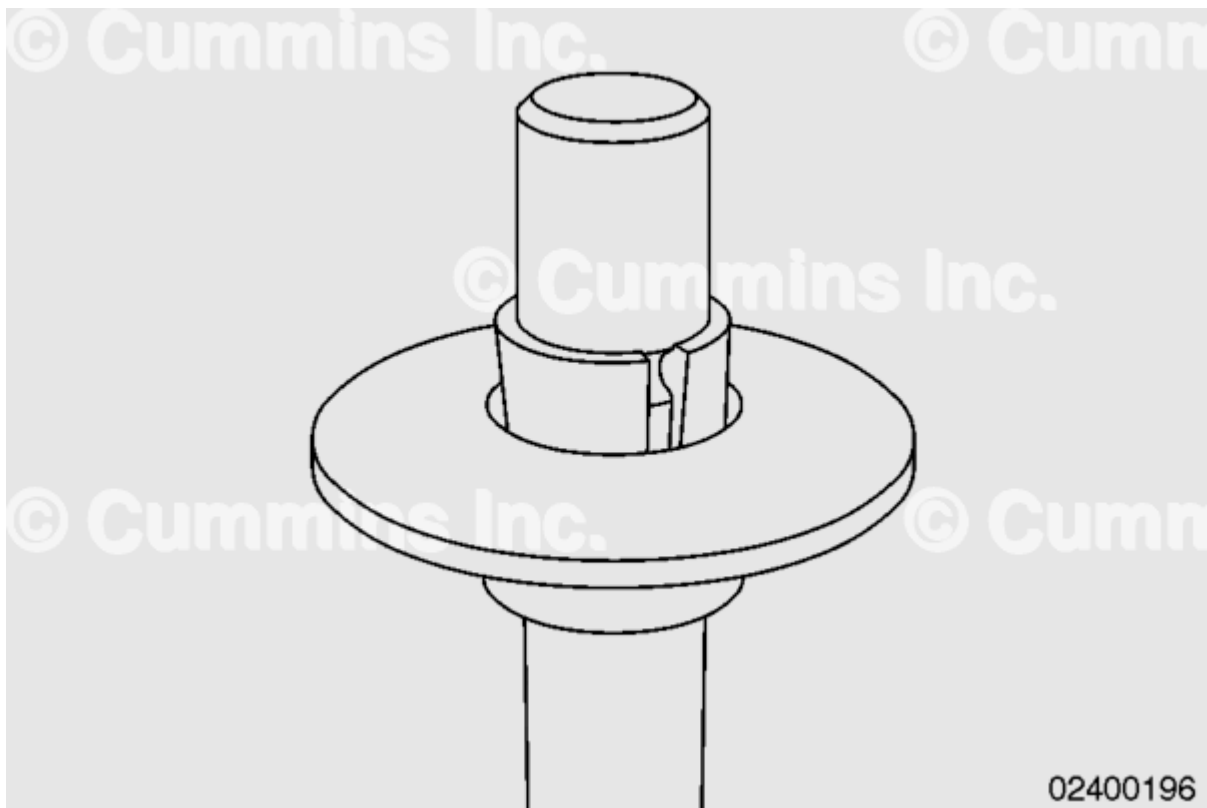
New retainer inside diameter at 2.3 mm [0.09 in] down from top: 13.14 mm [0.52 in]



Correct Installation (New Valve, Collets, and Retainer)



Incorrect Installation (New Valve and Retainer, Wrong Collets)



Incorrect Installation (New Valve and Collets, Wrong Retainer)

Note : It is extremely difficult to fit the new valve with the old collets and old retainer, as the retainer and collets will **not** seat at all.

Document History

Date	Details
2013-1-24	This revision is made in response to QSOL Online Helpdesk ticket #82884 (W/O 45474).

Last Modified: 29-Jan-2013
